

Module – 4

1-Resource Monitoring Techniques

Ans - Core Monitoring Techniques

- **Virtual Machine Monitoring:** It measures CPU utilization, RAM usage, and disk to ensure instances aren't over-provisioned or under-provisioned (means crashes).
- **Database Monitoring:** Focuses on performance, connection counts, and data integrity. It helps identify queries that might be bottlenecking an entire application.
- **Network Monitoring:** Analyzes the flow of data between services. Key metrics include latency, bandwidth usage, and packet loss.
- **Storage Monitoring:** Monitors the input/output operations per second (IOPS) and available capacity in cloud buckets (AWS S3) or Azure Disk.

2-How to access compute (windows and Linux) from internet? describe tools and its security

Ans - Windows Command Prompt : A command-line interface that can be used to access and manage remote Linux servers. It supports SSH and is a common tool for IT professionals.

MobaXterm: A terminal emulator that supports various network protocols, including SSH. It is lightweight and easy to use, making it suitable for users of all skill level. These tools provide a secure way to connect to remote servers, ensuring that data exchanged between the client and the server is encrypted and protected from unauthorized access. SSH is the primary protocol used for remote access, offering end-to-end data encryption and supporting advanced authentication methods.

3-Encryption Technologies and Methods

Ans - Symmetric Encryption -This method uses a single key for both encryption and decryption. It is efficient for large datasets but requires secure key sharing. Common algorithms include AES (Advanced Encryption Standard) and Triple Data Encryption Standard.

Asymmetric Encryption -Asymmetric encryption employs a pair of keys: a public key for encryption and a private key for decryption. It is widely used for secure communications and key exchanges.

Server-Side Encryption-This occurs after data is uploaded to the cloud but before it is written to disk. The cloud provider manages the encryption keys, ensuring compliance with many regulatory standards.

End-to-End Encryption-This ensures that only the sender and receiver can access the data. It is commonly used in secure messaging and file-sharing applications.

4-Describe network security in cloud, compute security and storage security

Ans -

- Virtual Private Cloud (VPC): Think of this as your private "bubble" within the public cloud. You use VPCs to isolate different parts of your business (e.g., HR data should never be on the same network as the public website).
- Security Groups (SGs): These act as a virtual firewall for your compute instances. In 2025, the best practice is Default Deny: all traffic is blocked unless you specifically write a rule to allow it.
- Web Application Firewalls : Specifically designed to stop Layer 7 attacks like SQL injection or Cross-Site Scripting that target your web apps.
- DDoS Protection: Cloud providers now offer massive protection (like AWS Shield or Azure DDoS Protection) that automatically absorbs "flood" attacks before they reach your servers.